

SOURCE-GROUNDED CNB TEST FIXTURE

Kraków Fast Tram Stage IV

Tram to Mistrzejowice

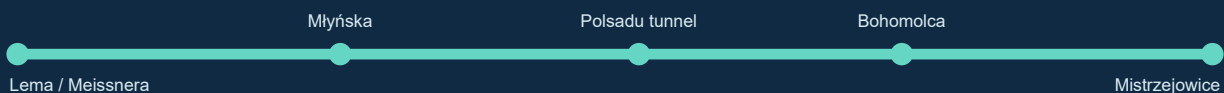
A real, late-construction sustainable urban mobility project assembled as a compact source document for testing the CityCatalyst Concept Note Builder.

City: Kraków, Poland

Sector: Sustainable transport, climate mitigation and urban resilience

Evidence cut-off: 17 August 2026

Document language: English



TESTING NOTICE

This is not an official City of Kraków concept note or funding application. It is a concise synthesis of public sources.

Source-backed facts, synthesis, proposed indicators and missing data are deliberately separated so the CNB flow can be tested for evidence discipline.

1. Project snapshot

Current status

Late construction and commissioning. On the evidence cut-off date, the project website displayed 99% design progress and 97% construction progress. A technical tram test at the connection to the existing network was completed successfully on 15 July 2026. The public target of opening by the end of Q2 2026 had already passed, and the reviewed sources did not confirm a revised revenue-service date. [S1] [S6] [S8]

Field	Source-backed project information
Project	Kraków Fast Tram Stage IV (KST IV), commonly presented as Tram to Mistrzejowice. [S1] [S2]
Public entity	City of Kraków, represented for the project by Zarząd Dróg Miasta Krakowa (ZDMK), the Kraków Municipal Road Authority. [S1]
Private partner	Consortium of PPP Solutions Polska and Gülermak. [S2] [S4]
Location	Northern Kraków, linking the Lema / Meissnera junction with the existing Mistrzejowice tram terminus through Prądnik Czerwony and Mistrzejowice. [S1] [S2]
Delivery model	Public-private partnership. The private partner finances, designs, secures permits, builds and then maintains the infrastructure for 20 years; the broader agreement is reported as 24 years. [S2]
Reported value	PLN 1.92 billion total project value, including approximately PLN 616 million net capital expenditure. These figures are not interchangeable and should remain separately labelled. [S2]
Financiers named	European Investment Bank (EIB), European Bank for Reconstruction and Development, Bank Pekao SA and Polish Development Fund. The EIB states that its financing agreement contributes PLN 180 million. [S2] [S4]

Why the project exists

- The northern districts served by the corridor are residential areas with offices, shops, extensive transport infrastructure and heavy traffic. [S2]
- The project connects two existing tram lines and supports Kraków's strategy to reduce central-city car traffic in favour of alternative transport. [S2] [S4]
- The new connection is intended to shorten the journey to the city centre by approximately 12 minutes and support tram headways of up to 90 seconds. [S1] [S2]

CNB-ready problem statement - synthesis

Residents of north-eastern Kraków need a faster, reliable and accessible alternative to private-car travel. The project closes a rapid-transit network gap while embedding green infrastructure in a dense urban corridor. This wording is a synthesis of public evidence, not a quoted official objective.

2. Scope, delivery and schedule



Component	Publicly documented scope	Evidence
Tram corridor	4.45 km route between Lema / Meissnera and Mistrzejowice.	[S2] [S4]
Tunnel	1,230 m tunnel built by open-cut method, including a multi-level interchange in the Polsadu area.	[S2]
Stops and terminus	Nine new tram stops, two underground, plus reconstruction of the existing Mistrzejowice terminus.	[S4]
Street and active travel	New or reconstructed tracks, stops, sidewalks, bicycle paths, roads and associated underground infrastructure.	[S1] [S2] [S5]
Green track	15,000 m ² of sedum was reported complete across designated surface-track areas in July 2026; an earlier design source described a 4.2 km single-track equivalent.	[S1] [S3]
Green stops and planting	Nineteen green-roof stop structures covering 1,900 m ² , replacement tree planting, 10,308 m ² of shrubs / flowerbeds / grasses, and a rain garden near Barei Roundabout.	[S3]
Systems and commissioning	Traction, switch control and heating, tunnel ventilation, SCADA monitoring, public address and passenger shelters. Commissioning was still underway in July 2026.	[S5] [S6]

Delivery sequence

Milestone	Evidence-based note
December 2021	PPP agreement signed between the City of Kraków and the private partner. [S2]
2023	Environmental reassessment received a positive opinion; construction began in July 2023. [S3]
October 2025	6,000 m ² of the planned 15,000 m ² of sedum had been installed. [S7]
July 2026	All designated surface green-track areas were reported complete; a network-connection technical test succeeded; tunnel systems and landscaping were still being commissioned. [S1] [S5] [S6]
17 August 2026	Website still displayed 97% construction and the passed end-Q2 opening target. Revised opening date requires confirmation. [S1] [S8]
Operations	Private partner to maintain the infrastructure for 20 years before city takeover. [S2]

3. Climate and development logic

Evidence discipline

The project is clearly framed by the EIB as sustainable public transport that can reduce carbon emissions and support lower car traffic. However, the reviewed public sources do not publish a project-specific tCO2e reduction estimate. CNB should preserve that distinction. [S4]

Climate mitigation pathway

- New rapid-transit capacity and shorter journeys make public transport more competitive with private cars. [S1] [S4]
- If the project produces a verified shift from car trips to tram trips, vehicle-kilometres and transport emissions should fall. This is a causal hypothesis to be measured, not a published result.
- Electric tram operations move tailpipe emissions out of the corridor. Net GHG benefit still depends on ridership, displaced car travel, tram electricity use, the electricity emission factor and embodied construction carbon.

Climate adaptation and local environment

- Sedum track is intended to retain rainfall, reduce surface heating, capture dust and reduce noise and vibration. [S7]
- Green-roof stop structures, replacement planting and the Barei rain garden add distributed green and blue infrastructure. [S3]
- The environmental design was reassessed by the Regional Directorate for Environmental Protection before construction. [S3]

Draft results framework for CNB testing

Level	Indicator	Known target / status	Evidence state
Output	Tram route delivered	4.45 km, including 1.23 km tunnel	Source-backed [S2] [S4]
Output	New stops delivered	9 stops, including 2 underground	Source-backed [S4]
Output	Green surface track	15,000 m² reported complete	Source-backed [S1]
Service	Journey-time saving	Approx. 12 minutes	Published forecast [S1] [S2]
Service	Peak service headway	Up to 90 seconds	Published design claim [S1]
Outcome	Annual passenger boardings	Baseline and target not found	Data gap
Outcome	Share of riders shifted from car	Baseline and target not found	Data gap
Impact	Net GHG emissions avoided	tCO2e/year not published in reviewed sources	Data gap - do not invent
Adaptation	Stormwater retained	Public source reports 4,750 litres/year for 1,900 m² green roofs; value and unit need verification	Source conflict / validation need [S3]

4. Beneficiaries, governance and financing

Stakeholder / beneficiary	Role or expected benefit
Residents of Prądnik Czerwony, Mistrzejowice and Olsza	Faster, more predictable access to the city centre and wider tram network; improved walking, cycling and interchange facilities.
People with reduced mobility	Accessible interchange and stop infrastructure are part of the public design description. Accessibility performance should be verified during commissioning.
City of Kraków / ZDMK	Public partner, contract manager and eventual infrastructure owner after the operating / maintenance term.
PPP Solutions Polska / Gülermak	Private partner responsible for financing, design, permits, construction and 20-year maintenance.
Public transport operator and network managers	Operate services and integrate traction, passenger information, interchange and road systems. Exact operating accountabilities were not detailed in the reviewed sources.
Residents and businesses along the corridor	Potential long-term accessibility and public-space gains, balanced against construction disruption, noise and access impacts.
Financiers	EIB, EBRD, Bank Pekao SA and PFR are named as co-financiers. EIB's stated contribution is PLN 180 million.

Financial interpretation

Keep three concepts separate

(1) PLN 1.92 billion is the reported total project value; (2) approximately PLN 616 million net is reported capital expenditure; (3) PLN 180 million is the EIB financing amount. CNB should not present any one of these as the complete funding gap.

[S2] [S4]

For a real funding concept note, the city should supply the payment mechanism, lifecycle-cost schedule, committed financing by lender, disbursement status, contingency, taxes, operating costs and any remaining financing need. None should be inferred from the three headline figures.

Implementation principles for a future concept note

- Treat commissioning and start of revenue service as the immediate delivery gate.
- Carry the PPP maintenance obligations into the results and monitoring plan, not only the construction narrative.
- Separate transport-service KPIs from climate-impact KPIs and assign data owners for both.
- Document accessibility, greenery survival and community disruption alongside ridership and travel-time results.

5. Risks, assumptions and missing evidence

Issue	Why it matters	CNB treatment
Opening-date slippage	The end-Q2 2026 public target had passed while construction was still reported at 97% and commissioning continued.	Flag as current schedule risk. Ask for approved revised date and critical-path status.
Construction vs. operational status	A successful technical run at the network tie-in does not prove the complete line is ready for passenger service.	Do not mark the project operational without acceptance and revenue-service evidence.
Modal-shift assumption	Climate benefit depends on who rides, what trip is displaced and service quality after opening.	Request baseline traffic, forecast ridership and ex-post passenger survey method.
GHG quantification	No reviewed source provides a project-level net tCO ₂ e estimate.	Require a transparent model covering displaced car travel, electricity and embodied carbon.
Cost interpretation	Total lifecycle project value, net capital expenditure and one lender's financing are different quantities.	Keep labels and currencies; request full financing plan before calculating a gap.
Green-infrastructure performance	Installed area is documented, but retention, temperature and biodiversity outcomes need operational monitoring.	Use survival, coverage, runoff and maintenance indicators with responsible owners.
Environmental and community impacts	Construction required vegetation removal, traffic changes and access disruption despite mitigation and new planting.	Include complaints, access, tree survival and restoration completion in monitoring.
Long-term PPP performance	Benefits depend on infrastructure availability and maintenance across 20 years.	Request availability standards, deductions, handback conditions and reporting cadence.

Minimum evidence to request from the city

- Approved opening / revenue-service date and remaining commissioning gates.
- Final route, stop and green-infrastructure as-built quantities.
- Ridership forecast, passenger baseline, traffic model and modal-shift assumptions.
- Project GHG methodology, electricity assumptions and embodied-carbon treatment.
- Complete sources-and-uses table, committed financing and lifecycle payment schedule.
- Environmental monitoring plan, tree / sedum maintenance plan and accessibility acceptance evidence.

6. Suggested CNB test scenarios

Recommended primary test prompt

Draft a climate-finance concept note for Kraków Fast Tram Stage IV using only this uploaded source. Separate documented facts from assumptions, preserve the current schedule caveat, and list the data needed before investment appraisal.

Expected extraction checks

Check	Expected CNB behaviour
Identity	Recognise KST IV / Tram to Mistrzejowice as the same project and name Kraków / ZDMK as the public side.
Quantities	Extract 4.45 km, 1.23 km tunnel, 9 new stops, 15,000 m² green track, approx. 12-minute saving and up-to-90-second headway with source labels.
Finance	Keep PLN 1.92 billion, PLN 616 million net capex and PLN 180 million EIB financing distinct.
Status	Report late construction / commissioning, not operational completion. Flag the stale opening target.
Climate	Explain the modal-shift and green-infrastructure logic without inventing a tCO2e result.
Gaps	Ask for ridership, modal split, emissions methodology, revised schedule, full financing plan and monitoring ownership.
Conflicts	Handle 4.5 vs 4.45 km as rounding; explain 24-year agreement vs 20-year maintenance; do not confuse 9 stops with 19 green-roof stop structures.

Useful follow-up prompts

- Which statements in the draft are source-backed, synthesized, proposed or missing?
- What evidence would be required to calculate annual net GHG reductions?
- Create a results framework that does not substitute construction outputs for climate outcomes.
- List schedule, financial, environmental, social and long-term maintenance risks.
- Show every place where the source is ambiguous or apparently stale.

Failure condition

The test should be considered failed if CNB invents an opening date, passenger forecast, funding gap, GHG reduction, or official city commitment that is not in this file.

7. Sources and provenance

All sources were accessed on 17 August 2026. Public project pages may change after this evidence cut-off. [S1], [S2], [S3], [S5], [S6] and [S7] are pages of the dedicated project information site; [S4] is the European Investment Bank; [S8] is the official City of Kraków website.

[S1] Tram to Mistrzejowice - project homepage and progress dashboard

<https://www.tramwajdomistrzejowic.pl/en/>

[S2] About - route, tunnel, project value, PPP structure and named financiers

<https://www.tramwajdomistrzejowic.pl/en/about/>

[S3] Tramway to Mistrzejowice with environmental agreements - planting, green track, green roofs and rain garden

<https://www.tramwajdomistrzejowic.pl/en/tramway-to-mistrzejowice-with-environmental-agreements/>

[S4] European Investment Bank - Poland: EIB supports the sustainable development of public transport in Krakow, 17 January 2023

<https://www.eib.org/en/press/all/2023-010-poland-the-eib-supports-the-sustainable-development-of-public-transport-in-krakow>

[S5] Construction Progress Update - 17 July 2026

<https://www.tramwajdomistrzejowic.pl/en/construction-progress-update-17-07/>

[S6] The first technical tram test run was completed successfully - 15 July 2026

<https://www.tramwajdomistrzejowic.pl/en/the-first-technical-tram-test-run-was-completed-successfully/>

[S7] The tram to Mistrzejowice is getting greener - 6,000 m² of sedum installed, 14 October 2025

<https://www.tramwajdomistrzejowic.pl/en/the-tram-to-mistrzejowice-is-getting-greener-6000-m%C2%B2-of-sedum-has-already-been-laid-on-the-tracks/>

[S8] City of Kraków - What will change for residents of Mistrzejowice and Prądnik Czerwony?, 29 July 2026

https://krakow.pl/dzielnica_xv_mistrzejowice/332364,2329,komunikat,co_zmieni_sie_dla_mieszkancow_mistrzejowic_i_pradnika_czerwonego_.html

Provenance note

This fixture paraphrases and organizes public information for testing. It intentionally does not reproduce source photography, logos or long quotations. Where a value appears questionable or stale, it is carried as a validation need instead of silently corrected.

Prepared for CityCatalyst Concept Note Builder testing.